

Five-Way Pac-Man

Cooperative Search Geometry for Automated Metalogical Deciders

Revision 003: certificate-backed settlement, conservative BANK simulation, and scheduling
geometry

Brian Tenneson
M.S., Applied Data Science
Currently Unaffiliated

September 19, 2026

This paper develops the five-Pioneer interpretation of AMLDs as a formal weighted state-space model. The Pac-Man language is mnemonic; all principal claims are stated for transition systems whose legal logical moves are generated by the inference rules of the underlying formal systems.

Copyright © 2026 Brian Tenneson. All Rights Reserved.

Abstract

An automated metalogical decider (AMLD) coordinates five specialized roles: prover P , refuter R , dependence certifier D , independence certifier I , and contradiction seeker C . This paper develops a “Five-Way Pac-Man” interpretation of that architecture. The formal object is not an arcade game but a typed, inference-rule-governed, weighted directed graph of complete AMLD states. Each accepted transition activates one Pioneer at a time; five families of settlement states play the role of exit classes; and the shared BANK can alter the locally available routes of several Pioneers even when only one Pioneer moves. Conservative BANK growth leaves the underlying object-theoretic consequence closure unchanged while allowing explicit-premise proof horizons to shorten, so a deposited lemma acts like opening a legal door rather than changing the underlying logic.

The paper defines the joint AMLD settlement horizon $H(X)$ as the minimum cost of a legal continuation from complete state X to any settlement class. For finitely settleable states with natural-valued transition costs, H satisfies a Bellman equation. A legal move $X \xrightarrow{a} Y$ is globally horizon-preserving exactly when $H(X) = c(a) + H(Y)$. The paper then proves the AMLD scheduling-regret identity: along every finite settlement run, total excess cost above the joint optimum is exactly the sum of the one-step horizon defects. This separates local Pioneer progress from global AMLD progress and makes precise the possibility that an optimal move can send the active Pioneer farther from its own exit while opening a much shorter route for another Pioneer. The resulting model provides a formal answer to the AMLD scheduling question: given one next unit of accepted computation, which Pioneer should move?

Contents

1	From five ATP roles to one cooperative maze	1
1.1	Complete AMLD state	1
2	Five exit classes and first settlement	3
3	The fixed global maze and the moving local walls	4
4	BANK deposits: opening doors without strengthening the object theory	4
5	Two floors: object search and metatheoretic search	5
6	Local Pioneer distance versus joint AMLD distance	5
7	The AMLD Bellman equation	6
8	The Five-Way Detour Principle	6
9	AMLD scheduling regret	7
9.1	Relation to classical shortest-path potentials	8
10	BANK jumps and global accounting	8
11	The scheduling problem: who should move next?	9
11.1	A round-robin baseline	9
12	Initial placement and history	10
13	Addressed roads: Hilbert coordinates without changing legality	10
14	A five-coordinate numeric compass	11

15 What Five-Way Pac-Man adds to the AMLD picture	12
16 Relation to induction	12
17 Boundaries of the model	13
18 Conclusion	13
Future direction	13

1 From five ATP roles to one cooperative maze

The AMLD architecture used here is inherited from the five-role formulation in *Depths of Induction, ML-Guided Proof Search, Cooperative AMLDs, and Formalized Self-Awareness, Version 48* [1]. Fix an object theory

$$F = (x, y, z),$$

using the Version 48 notation. Only the formula space y and inference-rule family z are used explicitly below. Fix an initial object BANK $B_0 \subseteq y$, a target $T \in y$, a designated refutation target $T^- \in y$, and a contradiction target $\perp \in y$. Fix also a declared metatheory F^{meta} capable of expressing the dependence and independence certificate languages actually used.

The standing Version 48 convention is crucial here. If $r \in z$ has arity k , then

$$r : y^k \rightarrow y$$

is a partial inference function. A logical proof step is legal exactly when its premise tuple lies in $\text{dom}(r)$, in which case the conclusion is the value of r . Thus no separate verifier primitive is used: legality is already carried by the inference rules of F (and, for metalogical steps, by the rules of F^{meta}).

The five AMLD roles are

$$\mathcal{J} := \{P, R, D, I, C\}.$$

They seek derivable settlement records of the forms

$$\begin{aligned} P &: B \vdash_F T, \\ R &: B \vdash_F T^-, \\ D &: \text{Dep}_{F,B}(T), \\ I &: \text{Ind}_{F,B}(T), \\ C &: B \vdash_F \perp. \end{aligned}$$

The P, R, C roles are object-theoretic; D, I are metatheoretic. The symbols $\text{Dep}_{F,B}(T)$ and $\text{Ind}_{F,B}(T)$ denote the protocol-declared metatheoretic certificate formulas. If the chosen dependence semantics treats an accepted proof or refutation as sufficient evidence of dependence, that implication is part of the declared certificate semantics; otherwise D remains separately certified. Thus dependence is not assumed here to be a fifth mutually exclusive object-level truth value.

The Five-Way Pac-Man picture is the following. There are five Pioneers, one for each role. They do not race on five independent boards. They occupy one joint state space. An “exit” is not necessarily a single terminal vertex but a member of one of five settlement classes. Only one Pioneer is activated per accepted logical tick. A move by one Pioneer can deposit a legally derived lemma into shared memory, thereby changing the routes explicitly available to other Pioneers.

There are no ghosts and no treasures in the model. In particular, there is no adversarial opponent. Mathematically the core object is a cooperative single-controller shortest-settlement problem on a rule-governed transition graph.

1.1 Complete AMLD state

Let S^J denote the local accepted search state of role $J \in \mathcal{J}$. Let B^{obj} and B^{meta} be type-distinct BANK components, let Π record certificate provenance, let $\Sigma \subseteq \mathcal{J}$ record accepted settlement classes, and let $t_{\text{set}} \in \mathbb{N} \cup \{\infty\}$ record the first-settlement stage (with ∞ before settlement). Let q contain whatever scheduler memory, learned-controller state, counters, or pseudorandom-generator state is needed to make the transition description Markovian. A complete accepted state has the form

$$X = (S^P, S^R, S^D, S^I, S^C, B^{\text{obj}}, B^{\text{meta}}, \Pi, \Sigma, t_{\text{set}}, q).$$

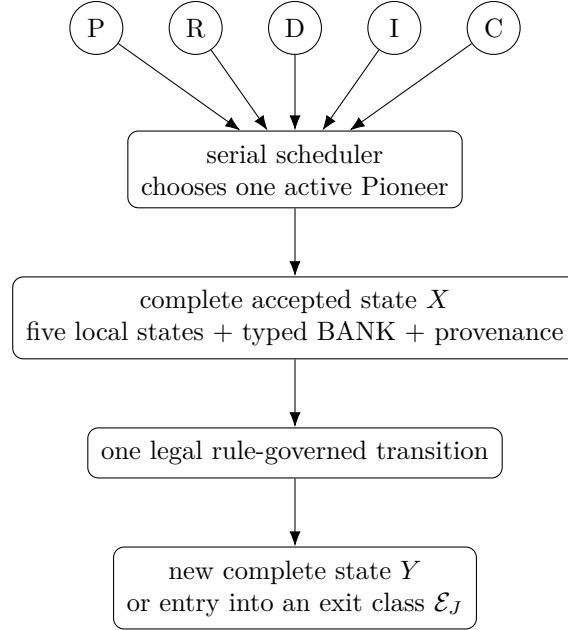


Figure 1: The Five-Way Pac-Man control picture. Five joysticks exist conceptually, but only one Pioneer is activated per accepted logical tick. The BANK is part of the complete state, so one Pioneer can change the routes available to the others.

This is the actual “position on the board.” Merely specifying the five local search positions is insufficient because the BANK, provenance, and controller state affect which subsequent transitions are legal and which action is selected.

Definition 1.1 (Certificate-backed settlement semantics). *For an accepted complete state X , a role may be added to $\Sigma(X)$ only when its declared certificate condition has been accepted with provenance in $\Pi(X)$:*

- (i) P : an F -derivation $B^{\text{obj}}(X) \vdash_F T$;
- (ii) R : an F -derivation $B^{\text{obj}}(X) \vdash_F T^-$ together with the protocol declaration that such a derivation is a sound refutation certificate for T (the special case $T^- = \neg T$ is allowed when the language contains that negation);
- (iii) D : an F^{meta} -derivation of the declared formula $\text{Dep}_{F,B}(T)$;
- (iv) I : an F^{meta} -derivation of the declared formula $\text{Ind}_{F,B}(T)$;
- (v) C : an F -derivation $B^{\text{obj}}(X) \vdash_F \perp$.

Settlement records are persistent: for every accepted transition $X \rightarrow Y$,

$$\Sigma(X) \subseteq \Sigma(Y).$$

The first-settlement coordinate t_{set} is set on the first transition for which Σ becomes nonempty and is thereafter unchanged. If a D - or I -certificate is formulated relative to a BANK that may later change, the protocol must either fix the referenced base or include a preservation theorem showing that accepted BANK updates preserve that certificate’s semantics.

Definition 1.2 (Serial AMLD action). *At any complete state X for which the protocol permits another accepted step, a serial AMLD action is a pair*

$$a = (J, m), \quad J \in \mathcal{J},$$

where J identifies the unique active Pioneer and m is one legal atomic local transition assigned to that Pioneer under the fixed AMLD protocol. An atomic transition may be a search step, inference step, certificate step, BANK update, or an inference together with its authorized BANK commit when the protocol explicitly packages those operations as one accepted transition. The resulting accepted transition is written

$$X \xrightarrow[a]{c(a)} Y,$$

where $c(a) \in \mathbb{N}^+$ is its declared cost.

The positive-integer convention is adopted for the main shortest-path theorems. It can be replaced by other well-founded additive cost sets when minima are known to exist.

Remark 1.3 (What serialization does and does not do). Allowing only one active Pioneer per logical tick provides an unambiguous accepted chronology and an unambiguous first-settlement time. It does not itself prove consistency. Logical trust comes from the inference rules of the object and metatheoretic formal systems together with the declared certificate semantics.

2 Five exit classes and first settlement

For each role J , let

$$\mathcal{E}_J := \{X : J \in \Sigma(X)\}$$

be the corresponding settlement class. Define the full settlement boundary

$$\mathcal{E} := \bigcup_{J \in \mathcal{J}} \mathcal{E}_J.$$

The classes need not be singleton sets, need not have the same cardinality, and need not be mutually exclusive. Some may be unreachable from a given initial state. Thus the informal phrase “five exits” means five *exit classes*, not literally five terminal squares.

Definition 2.1 (First-settlement game). *The first-settlement game beginning at X_0 stops at the least n for which $X_n \in \mathcal{E}$. Its cost is*

$$C(\rho) = \sum_{t=0}^{n-1} c(a_t)$$

for the run

$$\rho : X_0 \xrightarrow{a_0} X_1 \xrightarrow{a_1} \dots \xrightarrow{a_{n-1}} X_n.$$

The AMLD may continue after first settlement in a refinement mode. For example, a D -certificate may settle dependence first, after which a later P -certificate identifies the proof side. The historical first-settlement time remains fixed while Σ grows.

Proposition 2.2 (First-settlement minimum across roles). *If t^J denotes the first stage at which the run enters $\mathcal{E}_J$, with value ∞ if it never does, then*

$$t_{\text{set}} = \min_{J \in \mathcal{J}} t^J.$$

Proof. This is immediate from $\mathcal{E} = \bigcup_J \mathcal{E}_J$ and the definition of first entry into the union. $\square$

3 The fixed global maze and the moving local walls

The Pac-Man metaphor appears to suggest that the maze changes whenever the BANK grows. There is a cleaner formal representation.

Definition 3.1 (AMLD settlement graph). *For a fixed AMLD protocol, let $G_{\mathcal{A}}$ be the weighted directed graph whose vertices are all accepted complete states reachable from some admissible initialized state and whose edges are all legal serial AMLD transitions*

$$X \xrightarrow[a]{c(a)} Y.$$

The global graph $G_{\mathcal{A}}$ is fixed once the protocol, formal languages, inference rules, cost model, and transition rules are fixed. BANK growth is represented by movement to a different vertex having a different BANK coordinate.

Proposition 3.2 (Door-opening representation). *Suppose an accepted transition deposits a legally derived object-level lemma L and changes the BANK coordinate from B to $B \cup \{L\}$. Any new local action that becomes enabled because L is explicitly available is represented in $G_{\mathcal{A}}$ as an outgoing edge of the new complete state. No modification of the global graph is required.*

Proof. Vertices already encode the full BANK coordinate and edges are defined from the complete source state. Therefore “availability after deposit” is state dependence of the outgoing-edge relation, not a mutation of the already-defined transition graph. $\square$

Thus a Pioneer can correctly experience the event as “a wall opened” while the global mathematician treats the episode as movement through a fixed state graph.

4 BANK deposits: opening doors without strengthening the object theory

Write

$$\text{Con}_F(B) := \{\varphi \in y : B \vdash_F \varphi\}.$$

An object-level deposit L at state X is conservative when

$$B^{\text{obj}}(X) \vdash_F L.$$

After the accepted deposit, the new object BANK is $B \cup \{L\}$.

Theorem 4.1 (Conservative BANK invariance). *Assume every change of the object BANK has the append-only form $B \mapsto B \cup \{L\}$ with $B \vdash_F L$. Then along every accepted AMLD run*

$$\text{Con}_F(B_k^{\text{obj}}) = \text{Con}_F(B_0).$$

Proof. This is the standard AMLD conservative-BANK invariant [1]. Induct on object-BANK updates. Every update is append-only and every new deposit already lies in the consequence closure of the preceding BANK, so adjoining it does not change that closure. Monotonicity and idempotence of consequence closure give the equality at each stage. $\square$

For P, R, C , define the explicit-premise proof horizons

$$h_P(B) := \|T\|_F(B), \quad h_R(B) := \|T^-\|_F(B), \quad h_C(B) := \|\perp\|_F(B),$$

where $\|\varphi\|_F(B)$ is the minimum number of accepted inference steps in an F -proof of φ from explicit premises B , with length 0 permitted when φ is already an explicit premise, and with value ∞ if no proof exists.

Theorem 4.2 (Local proof-horizon monotonicity). *If $B \subseteq B'$, then for $J \in \{P, R, C\}$,*

$$h_J(B') \leq h_J(B).$$

Proof. Every proof from B is still a proof from B' . Taking the minimum over the larger set of available proofs cannot increase the minimum. $\square$

The two preceding theorems capture the door metaphor precisely:

consequence closure unchanged; explicit access may become shorter.

A deposited lemma can act as a macro or key. It can shorten a route measured from the current BANK without creating a new underlying theorem.

5 Two floors: object search and metatheoretic search

The maze is typed. It is useful to picture P, R, C as primarily occupying an object-theory floor and D, I as primarily occupying a metatheory floor. The metaphor must not erase the typing discipline.

A metatheoretic record such as

$$\text{Ind}_{F,B}(T)$$

is not automatically an object-level premise of F . Likewise, timeout or failure of P and R is not an I -exit. An I -settlement requires a derivation of the declared independence certificate in the metatheory. The same applies to a D -certificate.

This gives a multiplex geometry: local transitions occur in different typed components, while typed interfaces allow information to affect the shared complete state. The five Pioneers cooperate through the complete state, not through an untyped common pile of formulas.

6 Local Pioneer distance versus joint AMLD distance

The central distinction of the Five-Way Pac-Man model is between a distance belonging to one role and the distance belonging to the AMLD as a whole.

Definition 6.1 (Role-restricted settlement horizon). *For $J \in \mathcal{J}$, define $D_J(X)$ to be the minimum total cost of a path from X to $\mathcal{E}_J$ using only actions whose active role is J , with $D_J(X) = \infty$ if no such path exists. Zero-length paths are allowed, so $D_J(X) = 0$ when $X \in \mathcal{E}_J$.*

This is a search-control quantity, not the same as the proof-length horizon $h_J(B)$. The latter measures certificate length from explicit premises for object channels; D_J measures the cheapest J -only accepted execution under the declared action cost model.

Definition 6.2 (Joint AMLD settlement horizon). *For a complete state X , define*

$$H(X) := \min\{C(\rho) : \rho \text{ is a finite legal continuation from } X \text{ to } \mathcal{E}\},$$

with $H(X) = \infty$ if no finite settlement continuation exists. The zero-length continuation is allowed, so $H(X) = 0$ for $X \in \mathcal{E}$.

Because the costs are positive integers, whenever the displayed set is nonempty it is a nonempty subset of $\mathbb{N}$ and therefore has a least element.

Proposition 6.3 (Joint horizon is bounded by every role-restricted horizon). *For every J ,*

$$H(X) \leq D_J(X).$$

Proof. Every J -only path to $\mathcal{E}_J$ is also an admissible joint AMLD path to $\mathcal{E}$. $\square$

This inequality formalizes the advantage of allowing five cooperating roles: the joint controller is free to settle by whichever role and whichever sequence of cross-role actions yields the least total cost.

7 The AMLD Bellman equation

The next theorem gives the basic optimality equation for the Five-Way Pac-Man model.

Theorem 7.1 (AMLD Bellman equation). *Let $X \notin \mathcal{E}$ be a state with $H(X) < \infty$. Let $\mathcal{A}(X)$ be the legal serial actions from X , with successor $\text{Next}(X, a)$. Then*

$$H(X) = \min_{a \in \mathcal{A}(X)} (c(a) + H(\text{Next}(X, a))),$$

where actions with $H(\text{Next}(X, a)) = \infty$ cannot attain the minimum. For $X \in \mathcal{E}$, $H(X) = 0$.

Proof. Choose an optimal finite settlement path from X . Its first action a has successor Y , and the remaining suffix has cost at least $H(Y)$, so

$$H(X) \geq c(a) + H(Y) \geq \min_{b \in \mathcal{A}(X)} (c(b) + H(\text{Next}(X, b))).$$

Conversely, for any action b with finite successor horizon, concatenate b with an optimal continuation from $\text{Next}(X, b)$ to obtain a settlement path of cost $c(b) + H(\text{Next}(X, b))$. Hence $H(X)$ is no larger than every such quantity. Combining the inequalities gives equality. $\square$

Definition 7.2 (Horizon-preserving move). *A legal action a from X to Y is horizon-preserving when*

$$H(X) = c(a) + H(Y).$$

Corollary 7.3 (Optimal first-move characterization). *For finite $H(X)$, a legal action $a : X \rightarrow Y$ begins at least one globally minimum-cost settlement route if and only if it is horizon-preserving.*

Proof. If equality holds, concatenate the action with an optimal continuation from Y . Conversely, the first action of any globally optimal route must satisfy equality by the Bellman equation. $\square$

Thus the rigorous version of “which joystick should move next?” is: choose any legal action that preserves the joint horizon.

8 The Five-Way Detour Principle

A role-local greedy rule would insist that whenever Pioneer J moves, its own distance D_J should decrease. AMLD cooperation does not justify that rule.

Proposition 8.1 (Global optimality does not imply local progress). *There exist finite rule-governed five-role settlement graphs and a horizon-preserving action by role P ,*

$$X \xrightarrow{P} Y,$$

such that

$$D_P(Y) > D_P(X)$$

while

$$H(Y) < H(X).$$

Proof. Use unit costs and the following finite transition fragment, with no other outgoing edges from the displayed states. There is a P -only branch

$$X \xrightarrow{P} P_1 \xrightarrow{P} P_2 \xrightarrow{P} E_P, \quad E_P \in \mathcal{E}_P,$$

of length 3. There is also a legal P -action

$$X \xrightarrow{P, L \text{ committed}} Y,$$

followed by a P -only branch

$$Y \xrightarrow{P} Q_1 \xrightarrow{P} Q_2 \xrightarrow{P} Q_3 \xrightarrow{P} E'_P, \quad E'_P \in \mathcal{E}_P,$$

of length 4, while the newly explicit lemma L enables the one-step refuter edge

$$Y \xrightarrow{R} E_R, \quad E_R \in \mathcal{E}_R.$$

The unused D, I, C role coordinates may be inert on this fragment. Such a fragment is realizable inside a rule-governed AMLD by taking the displayed edges as protocol-authorized accepted transitions and assigning the L -commit its accepted provenance.

Therefore

$$D_P(X) = 3, \quad D_P(Y) = 4,$$

while

$$H(Y) = 1, \quad H(X) = \min\{3, 1 + 1\} = 2.$$

Thus $X \rightarrow Y$ is horizon-preserving because $H(X) = 1 + H(Y)$, even though $D_P(Y) > D_P(X)$. $\square$

Remark 8.2 (Why the construction is AMLD-shaped). The only cross-role mechanism used in the construction is a legal object-theory derivation whose BANK deposit changes the successor state's explicitly available transitions. Both the helping and helped channels remain on the object-theory floor, so no metatheory-to-object coercion is assumed.

This motivates the principle

Pioneer-local progress and AMLD-global progress are different quantities.

The active Pioneer may rationally spend effort on a lemma that is useless or even locally costly for its own settlement class if the deposit opens a shorter route for another role.

9 AMLD scheduling regret

The joint horizon allows every realized decision to be assigned an exact one-step defect from global optimality.

Definition 9.1 (One-step scheduling regret). *For a legal transition $X \xrightarrow[c]{c(a)} Y$ with finite $H(X)$ and $H(Y)$, define*

$$\delta(X, a, Y) := c(a) + H(Y) - H(X).$$

Proposition 9.2 (Nonnegativity and zero-regret characterization). *For every such transition,*

$$\delta(X, a, Y) \geq 0.$$

Moreover,

$$\delta(X, a, Y) = 0$$

if and only if the action is horizon-preserving.

Proof. If $X \notin \mathcal{E}$, the Bellman equation gives $H(X) \leq c(a) + H(Y)$. If $X \in \mathcal{E}$, then $H(X) = 0$ and $c(a) + H(Y) \geq 0$. Equality is exactly the definition of a horizon-preserving action. $\square$

The local defects telescope over a completed run.

Theorem 9.3 (AMLD scheduling-regret identity). *Let*

$$\rho : X_0 \xrightarrow{a_0} X_1 \xrightarrow{a_1} \cdots \xrightarrow{a_{n-1}} X_n$$

be a finite run with $X_n \in \mathcal{E}$. Then every $H(X_t)$ is finite and

$$C(\rho) - H(X_0) = \sum_{t=0}^{n-1} \delta(X_t, a_t, X_{t+1}).$$

Proof. Because the suffix from every X_t reaches $X_n \in \mathcal{E}$, each $H(X_t)$ is finite. Expanding the right side gives

$$\sum_{t=0}^{n-1} c_t + \sum_{t=0}^{n-1} H(X_{t+1}) - \sum_{t=0}^{n-1} H(X_t).$$

The horizon terms telescope to $H(X_n) - H(X_0)$. Since $X_n \in \mathcal{E}$, $H(X_n) = 0$. The expression therefore equals $C(\rho) - H(X_0)$. $\square$

Corollary 9.4 (Zero-regret run). *A finite settlement run is globally minimum-cost if and only if every transition on it has zero one-step scheduling regret.*

Proof. All one-step regrets are nonnegative. Their sum is zero exactly when each is zero, and by the regret identity the sum is zero exactly when $C(\rho) = H(X_0)$. $\square$

This is the central quantitative result of the paper. Every unit of excess settlement cost is exactly accounted for by local occasions on which the chosen action failed to preserve the joint settlement horizon.

9.1 Relation to classical shortest-path potentials

The Bellman recurrence itself is classical dynamic programming [4]. Likewise,

$$\delta(X, a, Y) = c(a) + H(Y) - H(X)$$

has the standard algebraic form of a reduced edge cost: under the convention $c_p(X, Y) = c(X, Y) + p(X) - p(Y)$, taking $p = -H$ gives exactly δ . The cancellation in the regret identity is the familiar telescoping of potential terms along a path [5]. The AMLD-specific content is therefore not a claim that Bellman optimality or potential telescoping is new. It is the specialization of those tools to a typed five-role state space with certificate-backed settlement classes, role-restricted horizons, conservative shared-BANK effects, and cross-Pioneer detours.

10 BANK jumps and global accounting

A BANK deposit can cause a dramatic drop in a channel-local explicit-premise proof horizon. That does not mean the joint optimum received the same amount of progress for free.

Theorem 10.1 (Endogenous breakthrough accounting). *Suppose a legal AMLD path of total cost a carries complete state X to complete state Y . Then*

$$H(X) \leq a + H(Y).$$

Equivalently, whenever both horizons are finite,

$$\boxed{H(X) - H(Y) \leq a.}$$

Proof. If $H(Y) = \infty$, the inequality is immediate in the extended nonnegative integers. If $H(Y) < \infty$, concatenate the given path from X to Y with an optimal settlement continuation from Y . This gives a legal settlement path from X of cost $a + H(Y)$. Since $H(X)$ is the minimum such cost, the inequality follows. $\square$

Consequently, when both complete-state horizons are finite, a one-step lemma deposit may reduce $h_P(B)$ by ten, one hundred, or more proof lines under a macro-sensitive explicit-premise metric, but it cannot reduce the complete-state joint horizon by more than the cost of the transition that was actually spent getting to the new state.

This resolves an apparent paradox in the door metaphor. Local doors can open dramatically. The global shortest path had already accounted for the possibility of walking to the key and then using the door.

11 The scheduling problem: who should move next?

An AMLD contains five local search processes, but under serial control only one receives the next accepted transition. Because scheduler memory and pseudorandom-generator state have been included in the complete state coordinate q , a deterministic state-feedback scheduler can be written simply as

$$\pi : X \mapsto a \in \mathcal{A}(X).$$

A genuinely stochastic formulation may instead use a transition kernel; the present paper keeps the Markov state representation so that the Bellman notation remains elementary.

The optimal policy, if exact H were available, would select

$$\pi^*(X) \in \arg \min_{a \in \mathcal{A}(X)} (c(a) + H(\text{Next}(X, a))).$$

This statement is descriptive, not an assumption that H is effectively computable. The AMLD formalism separates the value of an optimal continuation from the algorithmic problem of identifying it.

11.1 A round-robin baseline

A simple scheduler activates

$$P, R, D, I, C, P, R, D, I, C, \dots$$

in strict rotation, skipping only roles with no legal action. This is fair but generally ignores cross-role value.

Proposition 11.1 (Five-way round-robin route-emulation bound). *Assume unit-cost actions, no earlier settlement, and a fixed J -only settlement route of m actions that remains available regardless of foreign actions. Assume further that on each activation of J , its local controller executes the next action of that fixed route. Under strict five-way round robin, settlement occurs within at most $5m$ global actions.*

Proof. Between successive activations of J there are at most four accepted activations of other roles, and skipping roles can only shorten that delay. Under the stated availability and route-following assumptions, the m prescribed J -actions are therefore executed within at most $5m$ global actions. $\square$

The result is intentionally modest. It does not say round robin is optimal, does not compare wall-clock resource replication, and does not account for beneficial BANK deposits. Its purpose is to provide a clean scheduling baseline against which adaptive policies can be evaluated.

12 Initial placement and history

The five Pioneers need not begin at neutral or symmetric positions. Let $\mathcal{H}$ denote permitted prehistory: previously verified lemmas, cached certificates, learned indices, prior target families, or other authorized state used by the initialization protocol. An initializer is a map

$$\iota : \mathcal{H} \mapsto X_0.$$

Two AMLDs with identical transition rules can therefore face different settlement difficulty: for some permitted histories one may have

$$H(\iota(\mathcal{H}_1)) \neq H(\iota(\mathcal{H}_2)).$$

This separates two performance questions:

- (i) how favorable was the initial complete state, and
- (ii) how efficiently did the scheduler act after initialization?

The scheduling-regret identity addresses the second question relative to the supplied initial state. It does not erase advantages or disadvantages already encoded in initialization.

13 Addressed roads: Hilbert coordinates without changing legality

The maze may be supplied with coordinates without changing which moves are legal. Let

$$h : y \longrightarrow A_0$$

be a computable one-to-one WFF addressing with partial inverse h^{-1} on the occupied address set $h[y]$. In the MAPQUEST convention, WFFs remain intrinsically zero-dimensional points; inference geometry appears at the arity of the rule.

For a k -ary inference rule

$$r : y^k \rightharpoonup y,$$

define its addressed form

$$r_h : A_0^k \rightharpoonup A_0$$

by

$$r_h(a_1, \dots, a_k) = h\left(r(h^{-1}(a_1), \dots, h^{-1}(a_k))\right)$$

whenever every premise address decodes and the resulting premise tuple lies in $\text{dom}(r)$. Thus

the address says where; the inference rule says whether the road exists.

The legal addressed inference space is naturally represented by the tagged coproduct

$$\mathcal{G}_h = \bigsqcup_{r \in z} (\{r\} \times \text{dom}(r_h)).$$

A natural-number tick mechanism may enumerate candidate rational/Hilbert coordinates in an effectively countable implementation [3]. A candidate coordinate that fails to decode to premises in $\text{dom}(r)$ is simply an unoccupied or blocked road; the inference-rule domain already determines that fact.

This is the interface between Five-Way Pac-Man and Omnicartography. Pac-Man describes actual legal movement of the five Pioneers. The Omnicartographer may use Hilbert, rational-cube, natural-number, or other reversible coordinates to map and compare those roads before selecting a direction.

14 A five-coordinate numeric compass

The earlier numeric AMLD development can be expressed directly in the five-Pioneer geometry [2]. Define, for $J \in \mathcal{J}$,

$$\mu_J(X) = \begin{cases} \frac{1}{1 + D_J(X)}, & D_J(X) < \infty, \\ 0, & D_J(X) = \infty, \end{cases}$$

and collect the role coordinates into

$$\boldsymbol{\mu}(X) = (\mu_P, \mu_R, \mu_D, \mu_I, \mu_C) \in [0, 1]^5.$$

Define also the exact joint settlement coordinate

$$\mu_*(X) = \begin{cases} \frac{1}{1 + H(X)}, & H(X) < \infty, \\ 0, & H(X) = \infty. \end{cases}$$

Proposition 14.1 (Numeric compass equivalence). *For states X, Y with finite joint horizons,*

$$H(X) < H(Y) \iff \mu_*(X) > \mu_*(Y).$$

For unit-cost transitions, a horizon-preserving move from a nonterminal state strictly increases μ_ .*

Proof. The map $d \mapsto 1/(1 + d)$ is strictly decreasing. If $X \rightarrow Y$ is horizon-preserving with unit cost, then $H(Y) = H(X) - 1$, so the transformed coordinate strictly increases. $\square$

The vector $\boldsymbol{\mu}$ is not a fuzzy truth assignment. It records five role-restricted navigation opportunities. In particular, two states may have the same μ_* while being close to different exit classes.

A conservative BANK lemma can also be given an access-effect profile. If $X \oplus L$ denotes a matched state differing from X only by making a legally derived lemma L explicitly available (and adding its provenance), define

$$\Delta_L(X) = \boldsymbol{\mu}(X \oplus L) - \boldsymbol{\mu}(X).$$

Definition 14.2 (Role-path preserving BANK extension). *The extension $X \mapsto X \oplus L$ is role-path preserving when, for every $J \in \mathcal{J}$, every finite J -only path from X to $\mathcal{E}_J$ has a J -only path from $X \oplus L$ to $\mathcal{E}_J$ of the same total cost. This is stronger than merely preserving the actions legal at the first state; it requires the old role-restricted route to remain reproducible through its successive BANK-extended states.*

Proposition 14.3 (Nonnegative BANK access profile). *If $X \mapsto X \oplus L$ is role-path preserving, then for every $J \in \mathcal{J}$,*

$$D_J(X \oplus L) \leq D_J(X),$$

and consequently every component of $\Delta_L(X)$ is nonnegative.

Proof. Every finite J -only settlement path available from X has a cost-preserving counterpart from $X \oplus L$. Taking minima gives $D_J(X \oplus L) \leq D_J(X)$, with the statement also valid when the old value is ∞ . Since $d \mapsto 1/(1+d)$ is decreasing on finite d and the convention maps ∞ to 0, the corresponding compass coordinate cannot decrease. $\square$

This coordinate isolates the shortcut value of explicit access; the actual cost of reaching and depositing L is still charged by the global accounting theorem.

15 What Five-Way Pac-Man adds to the AMLD picture

The model supports several conclusions that are easy to miss when an AMLD is described merely as five ATP-like channels running in parallel.

1. **The unit of geometry is the complete AMLD state.** Five local search positions alone do not determine the future; BANK contents, provenance, typing, and settlement records matter.
2. **The five roles are cooperative, not independent.** One role can deposit a verified object lemma that changes another role’s explicitly available route.
3. **BANK growth can change access without changing deductive strength.** Under conservative deposits, the consequence closure remains fixed while explicit-premise horizons can shrink.
4. **The relevant global objective is first settlement, not local monotone progress.** A move can be globally optimal even when the active Pioneer becomes farther from its own exit class.
5. **Serial scheduling is a mathematical control problem.** The next decision is not only “what should P try?” but “which of P, R, D, I, C should receive the next accepted unit of computation, and what should that role do?”
6. **Excess settlement cost has an exact decomposition.** The scheduling-regret identity expresses total excess cost over the joint optimum as a sum of one-step horizon defects.

These points distinguish an AMLD from five unrelated ATPs. The shared BANK and the common first-settlement objective turn the five local searches into one coupled state-space process.

16 Relation to induction

Nothing in the Five-Way Pac-Man formalization introduces a new induction axiom. The complete AMLD chronology

$$X_0, X_1, X_2, \dots$$

supports ordinary induction on accepted global stage. BANK birth-time arguments use induction on natural-number birth stages together with finite induction over certificate DAGs. Structural and rule induction remain those of the underlying object and metatheoretic languages.

The contribution developed here is geometric and decision-theoretic: the same rule-governed chronology is viewed as a weighted path in a joint state graph, permitting settlement horizons, Bellman optimality, detour phenomena, and exact scheduling regret to be defined and proved.

17 Boundaries of the model

Several limitations are deliberate.

First, $H(X)$ is a mathematical optimum. This paper does not assert that an ordinary effective AMLD can compute H uniformly. The name “automated metalogical decider” denotes the five-role architecture used here; it does not assert total termination or decidability for arbitrary inputs. Second, a shorter explicit-premise proof horizon need not imply shorter wall-clock search. Additional BANK material can increase branching or indexing costs. Third, no independence result follows from failed proof search. Fourth, the state graph may be enormous or infinite; the graph semantics is a definition of the process, not a claim that the whole graph can be explicitly stored. Fifth, the paper treats serial accepted transitions. Parallel implementations require an additional concurrency semantics before the same first-settlement and regret formulas can be applied without modification.

These boundaries are useful because they isolate the contribution: a rigorous cooperative geometry for the existing AMLD architecture, without requiring an ideal Oracle or complete Omnicartographer assumption.

18 Conclusion

The Five-Way Pac-Man interpretation can be summarized as

five specialized Pioneers + one typed BANK + one accepted chronology + five exit classes.

Formally this is a weighted directed graph of complete rule-governed AMLD states. The BANK gives the five roles a shared environment: a move by one Pioneer can alter the explicitly available routes of the others while conservative deposits preserve the underlying object-theoretic consequence closure.

The central distance is therefore not any single role’s distance to its own settlement class but the joint horizon

$$H(X) = \text{minimum legal cost from the complete state } X \text{ to any settlement class.}$$

It obeys the Bellman equation, characterizes globally optimal joystick moves, permits locally adverse but globally optimal detours, and yields the exact identity

$$C(\rho) - H(X_0) = \sum_t \delta_t.$$

Thus the fundamental AMLD control question becomes precise: with one next accepted unit of computation available, which Pioneer should move? Five-Way Pac-Man recasts AMLD scheduling as the search for low-regret control of five cooperating logical explorers rather than as five independent theorem-proving races.

Future direction

The natural companion is the Omnicartographer: a controller with additional map information that can use addressed inference geometry, bounded counterfactual look-ahead, and settlement-horizon estimates to select among the legal roads. The present paper establishes the five-Pioneer movement geometry and value function that such cartographic guidance acts upon.

References

- [1] B. Tenneson, *Depths of Induction, ML-Guided Proof Search, Cooperative AMLDs, and Formalized Self-Awareness, Version 48*, September 17, 2026.
- [2] B. Tenneson, *Numeric AMLD: Fuzzy-Looking Settlement Coordinates with Binary Verifier Semantics*, September 2026. The present revision retains the distance-coordinate mathematics while replacing verifier-as-primitive language by the Version 48 inference-rule semantics.
- [3] B. Tenneson, *Natural-Number Tick Addressing for SIC-Executable Hilbert Proof Navigation, Working Algorithm 001*, September 9, 2026.
- [4] R. Bellman, *Dynamic Programming*, Princeton University Press, 1957.
- [5] MIT OpenCourseWare, *6.006 Introduction to Algorithms, Lecture 14: Johnson's Algorithm*, Spring 2020, <https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/>.